Experiment – 9
Continuous-time LTI system

/MATLAB Drive/Experiment_9_1.m

```
clc; clear; close all;
% Continuous-Time LTI System: dy(t)/dt + 2y(t) = x(t)
% Assume zero initial conditions: H(s) = Y(s)/X(s) = 1/(s + 2)
num = [1];
den = [1 2];
sys = tf(num, den);
disp('Transfer function:');
sys
disp('Pole:');
disp(pole(sys));
disp('Zeros:');
disp(zero(sys));
t = 0:0.01:5;
figure;
impulse(sys, t);
grid on;
title('Impulse Response');
xlabel('Time (s)');
ylabel('h(t)');
figure;
step(sys, t);
grid on;
title('Step Response');
xlabel('Time (s)');
ylabel('y(t)');
% Response to x(t) = sin(2t)u(t)
x = sin(2*t);
figure;
lsim(sys, x, t);
grid on;
title('Response to sin(2t)u(t)');
xlabel('Time (s)');
ylabel('y(t)');
```

Code

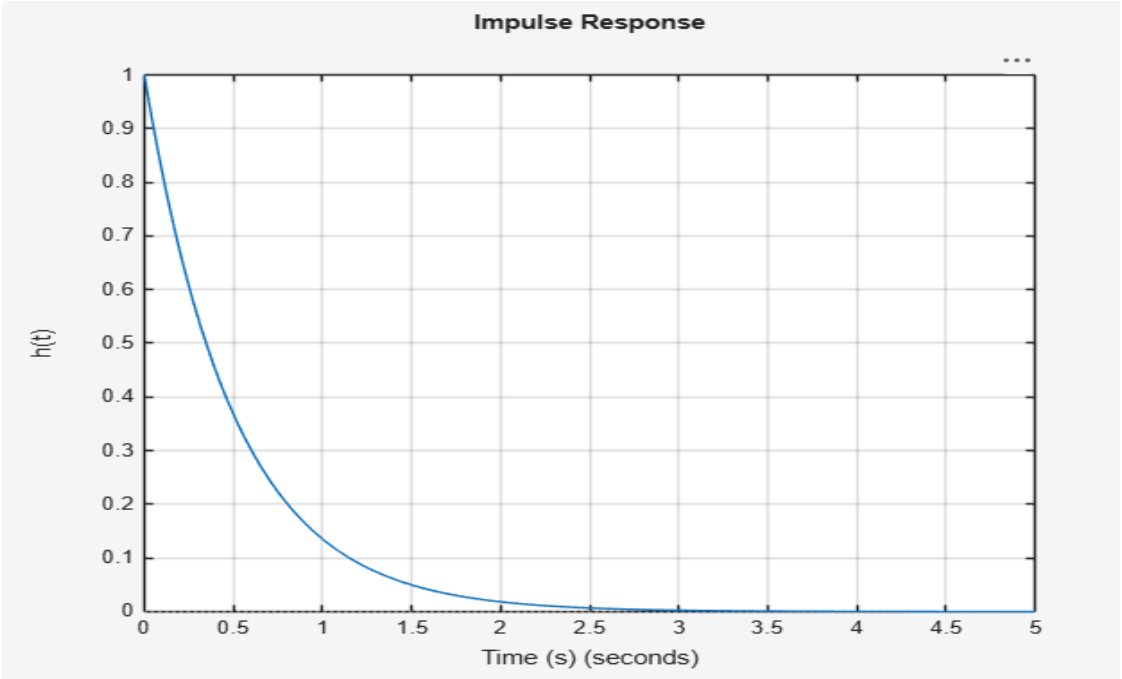


Figure 1

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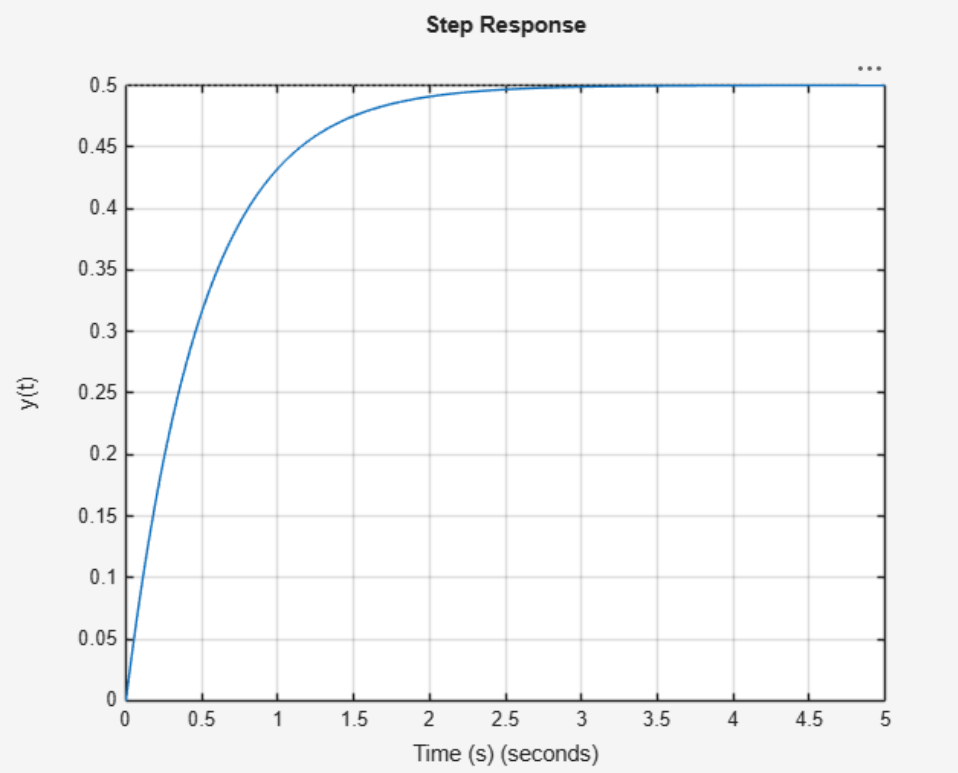


Figure 2

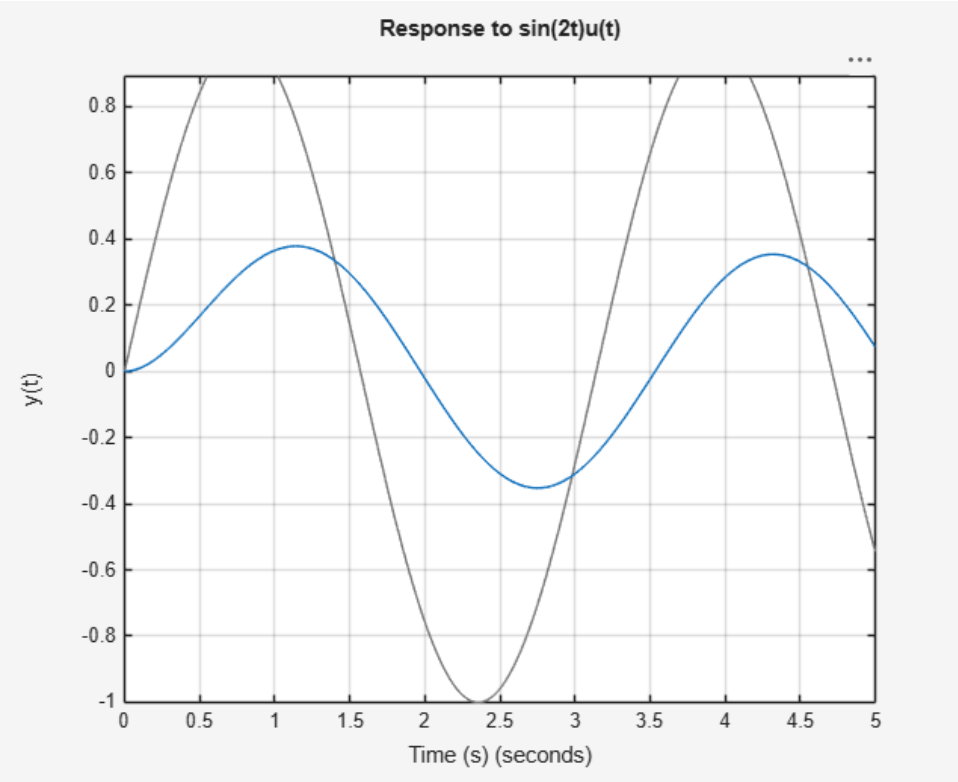


Figure 3